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Virtual Screening of Natural Flavonoids against Cyclin-Dependent Kinase 2 in Cancer Therapy

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Virtual Screening of Natural Flavonoids Against Cyclin-Dependent Kinase 2 in Cancer Therapy

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1 Introduction

Cancer is characterized by uncontrolled cellular proliferation and evasion of cell death. This excessive stem-cell proliferation arises from cell-cycle machinery failure, which ordinarily safeguards genomic stability. (Almalki, 2023). Cyclin-dependent kinases (CDKs) regulate this machinery. CDK dysfunction disrupts the cell cycle and facilitates malignant transformation and tumor development. (Siddique et al., 2024). Ultimately, this dysregulation disrupts cellular homeostasis (Hsu et al., 2025). Cyclin-dependent kinase 2 (CDK2) serves as a critical regulator of the G1-to-S-phase transition, promoting cell-cycle progression by initiating DNA replication (Davies et al., 2002). Clinically, CDK2 overexpression contributes to aggressive cancer growth and mortality in high-grade breast and ovarian cancers. (Taheri et al., 2026). Type 1 inhibitors target the ATP-binding pocket in the catalytic cleft of CDK2's N- and C-terminal domains. (Elmasry, 2023). ATP-site targeting may inhibit kinase-mediated phosphorylation (House et al., 2025). Resistance to CDK4/6 inhibition may involve CDK2-pathway adaptation. Cyclin E amplification and upregulation may enable tumor cells to evade G1-phase arrest and sustain proliferation despite CDK4/6 inhibition. This resistance loop underscores the need for intervention (Stachyra et al., 2026). Natural flavonoids may modulate hyperactive oncogenic pathways without apparent cellular damage (Tavsan & Kayali, 2019). These compounds may inhibit ATP binding at

the ATP-binding pocket, downregulate oncogenic drivers, and upregulate endogenous tumor suppressors (Stachyra et al., 2026; Wang et al., 2025).

Cyclin-dependent kinase 2 (CDK2) is a serine/threonine kinase that regulates the G1/S phase of the cell cycle. Early studies showed that CDK2 has a typical bilobal structure of protein kinases; the region surrounding the catalytic site is not properly arranged for efficient substrate binding (Asghar et al., 2015). CDK2 activation depends strongly on its interaction with cyclin. Analysis of the structure of cyclin A-CDK2 shows that binding produces major conformational changes, and the activation loop helps reposition important residues and create a more favorable active site configuration (De Azevedo et al., 1996). Cyclin binding not only regulates CDK2 but also functions as an important structural switch that may promote kinase activation. Full activation is controlled by phosphorylation of Thr15 and Thr14, providing an inhibitory mechanism for controlling CDK2 activity. Chemical-genetic studies demonstrate this (Chao & Price, 2001; Whittaker et al., 2017). Studies in human cells demonstrate that CDK7 supports efficient CDK2 activation (Senderowicz et al., 1998). After activation, its substrate specificity is partly determined by interaction with the cyclin complex or related cyclin-binding motifs present in target proteins. These interactions position target proteins for phosphorylation (Woyach et al., 2012). Cyclin E1-CDK2 studies show structural features that distinguish this complex from cyclin A-CDK2 (Meijer et al., 1997). Abnormal shifts in cyclin E-CDK2 are particularly relevant to cancer because they shift the cells toward proliferation. At the cellular level, CDK2 activity is not shown to be a simple on-or-off process; it has been shown that differences in CDK2 activity after mitotic exit can influence whether a cell can continue proliferating or enter a quiescent state (Bach et al., 2005). Evidence from genetics indicates that CDK2 function is highly context-dependent (Benson et al., 2007). In oncogenic cells, CDK2 activity can be suppressed by c-MYC; however, CDK2 can help maintain proliferation under oncogenic conditions (Le Tourneau et al., 2010). Links between abnormal cyclin and CDK2-dependent tumorigenesis come from studies of the low molecular weight of cyclin E; these tumor-associated cyclin E isoforms have increased activity for abnormal cell progression (Fry et al., 2004). Animal models reveal their importance, where the low molecular weight cyclin E promoted mammary tumor and metastatic disease (Losiewicz et al., 1994). This tumor-promoting effect depends on CDK2; loss of CDK2 substantially prevented mammary tumor development, providing evidence that CDK2 is required for these specific oncogenic pathways (Carlson et al., 1996). In ovarian cancer, the role of cyclin E has also been

shown (Senderowicz, 1999). Specifically, the LMW form is relevant to patient survival (Keyomarsi et al., 2002). CDK2 contributes to persistent cell proliferation, genomic instability, tumor progression, and therapeutic resistance in cancer settings. Flavonoid investigation may offer a means of attenuating these mechanisms and potentially limiting therapeutic resistance, as recent studies show that flavonoids can interact with the binding pocket of CDK2 (De Azevedo et al., 1996).

Deregulated CDK2 activity may be a major driver of cancer. The first CDK2 inhibitor was developed in the 1990s, Researchers anticipated direct therapeutic translation of cell-cycle regulation. (Asghar et al., 2015). The early agent showed that the CDK ATP site is druggable, but the pan-CDK inhibition, without mechanistic clarity or protein selection, may be clinically insufficient. Chemical optimization produced PD 0332991 palbacilib. It inhibits CDK4 (1C50 0.011 µM) and CDK6 (0.016 µM), shows no effect against 36 other kinases, and induces G1 arrest exclusively in Rb-positive cells. (Kaur et al., 1992). Natural flavonoids typically exhibit low bioavailability, poor absorption, and rapid metabolic degradation. (Thilakarathna & Vasantha Rupasinghe, 2013). To address these limitations, the docked compounds were subsequently evaluated using absorption, distribution, metabolism, excretion, and toxicity (ADMET) filters and molecular-dynamics simulations.

2. Materials and Methods

2.1 Protein structure retrieval and preparation

The 3D structure of human cyclin-dependent kinase 2 (CDK2) protein was retrieved the from the RCSB Protein Data Bank. (Rose et al., 2017). X-ray diffraction determined structure 4KD1 at 1.70 Å resolution; it contains one CDK2 chain (chain A) with 298 residues. (Martin & Schonbrunn, 2013). For the structural analysis and molecular docking, the WSL Ubuntu Linux system was used by following the GitHub repository (https://github.com/biochemi/pdb_extract). The downloaded structure was further processed before molecular docking, which was performed by following all the commands in the repository. Then the structure was further evaluated for structural cleaning and preparation for docking. Then the file was exported as a cleaned PDB and converted to pdbqt format for the required AutoDock Vina. (Eberhardt et al., 2021; Trott & Olson, 2010).

2.2 Ligand retrieval and preparation

For ligand retrieval, a virtual screening library of flavonoids was prepared for the subsequent molecular docking studies against CDK2. Initially, 100 flavonoids were selected as ligand compounds and processed using the GitHub repository (https://github.com/biochemi/pubchem_extract_optimize), based on the ligand PubChem preparation workflow. The computational ligand preparation pipeline was implemented in Python and included PubChem retrieval, RDKit-based structure optimization, Open Babel conversion, and automated validation of the generated ligand library. The names of the compounds were updated and saved successfully into the compound name of the retrieved GitHub for the retrieval of the CIDs through the PUG REST. (S. Kim et al., 2018). An interface for each compound, preferably a 3D structure, was retrieved from PubChem. If a 3D structure was unavailable, the pipeline was designed to download a 2D structure. (Bolton et al., 2011). Further, the retrieved structure was organized into a numbered SDF library. The retrieved ligand structure was subjected to the geometry optimization pipeline supported by the MMFF94, MMFF94s, UFF, and ETKDG methods using RDKit; out of the 100 compounds, 97 were available for optimization.

The optimized structure was further subjected to additional ligand quality control using RDKit, in which MW, LogP, HBD, HBA, RotB, TPSA, and Lipinski's rule of five (RO5) were used to to categorized ligand according to commonly MW $\leq$500 Da, calculated Logp ≤ 5, HBD ≤ 5, and

HBA ≤ 10. (Lipinski et al., 1997). Based on those criteria, 72 compounds satisfied the criteria, and 25 compounds failed, violating one or two of the rules. The chloride counterions were removed while retaining the corresponding flavonoids; duplicate structures were also removed. Then they were finally cleaned and validated; the cleaned library contains 95 remaining structures. Then the structures were combined into a single library. For the preparation Autodock vina based docking hydrogen atom added, assigned Gasteinger charges, specifically for vina then converted in pdbqt using Open Babel version 3.1.1 (O'Boyle et al., 2011).

2.3 Identification of active site of the receptor

An active site is a specific area found on the chain of the receptor where the compounds tend to bind and cause conformational changes to induce pharmaceutical activities. (Bongrand, 1999). One more important part of molecular docking simulation is the determination of the binding area of the ligand. An active site specifies the drug binding site. (Hussein et al., 2015; Janave et al., 1999).

2.4 Grid box determination

The docking site was defined from the binding site of the co-crystallized ligand in the experimental structure of the CDK2 receptor (4KD1). The docking box for 4KD1 was centered at x = 55.584 Å, y = 79.685 Å, z = 28.275 Å, with a docking box size of 18 × 18 × 18 Å.

2.5 Molecular Docking and Protein-Ligand Interaction

Following preparation and optimization of the protein and selected ligand, AutoDock Vina was used for the prediction of protein-ligand interaction poses and binding energy (Eberhardt et al., 2021; Trott & Olson, 2010). The top 10 docking poses in complex with the receptor were subsequently analyzed by characterizing hydrogen bonds, hydrophobic contacts, and salt bridges, as well as the interacting residues using the Protein-Ligand Interaction Profiler (PLIP) (Adasme et al., 2021; Salentin et al., 2015). BIOVIA Discovery Studio (*T101-2024 Notification Regarding BIOVIA Discovery Studio 2025*, n.d.) was used to further generate the three-dimensional structure of the binding pocket, and a 2D structure image was generated.

2.6 ADMET Analysis

For the computational determination of absorption, distribution, metabolism, excretion, and toxicity in this study, the requirements given below were analyzed.

GI absorption

BBB permeability

CYP inhibition

Hepatotoxicity

Mutagenicity

Skin sensitization

Drug likeness

ADMETlab 3.0 (Fu et al., 2024). was used to obtain the requested parameters and computational predictions for the ligands. For LD50 analysis, Protox 3.0 (Banerjee et al., 2024). was used to assess the acute oral toxicity of the selected docked ligand.

2.7 Molecular dynamics Simulation

Molecular dynamics simulations were performed using GROMACS. (Abraham et al., 2015). To investigate the stability of the docked complex ligand under the Ubuntu Linux system. The AMBER99SB-ILDN force field (Lindorff-Larsen et al., 2010). Was used to describe the protein. The TIP3P water model was employed for solvation of the system. (Jorgensen et al., 1983). The ligand topology Antechamber Python parser interface (ACPYPE) tool was used. (Sousa Da Silva & Vranken, 2012).

3. Results

3.1 Molecular docking

Molecular docking scores for the ten highest-ranked complexes are reported in Table 2. For residue-level interaction analysis, the ten highest-ranked compounds docked in complex with 4KD1 were assessed for hydrogen bonds, hydrophobic interactions, and salt bridges; residue numbers, interaction distances, and interaction counts are reported in Table 1. The three-dimensional protein–ligand complex and its two-dimensional structure are shown in Fig. 1.

Table 1

Hydrogen Bond, Hydrophobic interaction, and salt bridge interacting residues

Ligand	Hydrogen Bond		Hydrophobic interaction		Salt Bridges	
	Amino acid residues	Distance (Å)	Amino acid residues	Distance (Å)	Amino acid	Distance (Å)
Hesperidin	GLU-8	3.02	VAL-18	3.5	LYS-20	3.78
	ILE-10	2.71	PHE-82	3.99		
	THR-14	3.92	ASP-86	3.93		
	LEU-83	3.16	ASN-132	3.74		
	LEU-83	3.38	LEU-134	3.81		
	ASP-86	3.88	LEU-134	3.87		
	ASP-86	2.49				
	LYS-89	2.79				
	GLN-131	3.06				
	LEU-298	3.77				
CorylifolA	THR-14	3.23	ILE-10	3.64		
	LEU-83	3.21	ILE-10	3.66		
	LYS-129	3.74	VAL-18	3.54		
	GLN-131	3.64	ALA-31	3.86		
			PHE-82	3.70		
			ASP-86	3.98		
			GLN-131	3.80		
			ASN-132	3.36		
			LEU-134	3.32		
Catechin gallate	LYS-33	3.57	ILE-10	3.57	LYS-89	4.40
	LYS-33	3.64	ILE-10	3.87		
	LYS-33	3.06	VAL-18	3.99		
	ASP-86	2.51	ALA-31	3.76		
	ASP-86	3.51	PHE-82	3.82		
	LYS-89	3.27	PHE-82	3.28		
			ASP-86	3.81		
			GLN-131	3.51		
			LEU-134	3.74		
			LEU-134	3.57		
Baicalin	ILE-10	3.51	ILE-10	3.74	LYS-33	3.48
	THR-14	3.51	ILE-10	3.45	LYS-89	5.34
	LYS-33	3.57	LEU-134	3.69		

	LEU-83	3.03	LEU-134	3.32		
	GLN-131	3.77				
Corylin	GLN-85	4.08	ILE-10	3.75		
			ILE-10	3.45		
			VAL-18	3.70		
			LYS-33	3.94		
			VAL-64	3.52		
			PHE-80	3.55		
			PHE-80	3.88		
			PHE-82	3.75		
			PHE-82	3.58		
			LYS-89	3.45		
			LEU-134	3.54		
			LEU-134	3.84		
			ALA-144	3.64		
Eriocitrin	GLU-8	3.04	ASP-86	3.77	LYS-89	3.51
	GLU-8	3.99	GLN-131	3.69	LYS-89	5.30
	THR-14	3.18	ASN-132	3.44		
	THR-14	3.86				
	LEU-83	4.05				
	ASP-86	3.11				
	LYS-129	3.77				
	LYS-86	3.77				
	LYS-129	3.77				
	GLN-131	3.84				
	ASN-132	3.24				
Weighteone	THR-14	3.41	ILE-10	3.86		
	LEU-83	3.12	ILE-10	3.48		
	ASP-86	2.88	PHE-82	3.67		
	ASP-86	2.88	GLN-131	3.89		
	LYS-129	3.80	ASN-132	3.61		
	GLN-131	3.44	LEU-134	3.58		
			LEU-134	3.70		
Lupiwighteone	THR-14	3.46	ILE-10	3.34		
	LYS-33	4.04	ALA-31	3.54		
	LEU-83	2.85	GLN-131	3.79		
	LEU-83	2.57	ASN-132	3.65		
	LYS-129	4.03	LEU-134	3.82		
	GLN-131	3.29				
Gallocatechin	LYS-33	3.62	ILE-10	3.67	LYS-89	4.31
gallate	GLU-81	3.52	ILE-10	3.85		
	LEU-83	2.80	VAL-18	3.83		
	ASP-86	2.74	PHE-82	3.22		
	LYS-89	3.33	ASP-86	3.76		
			GLN-131	3.50		
			LEU-134	3.90		
			LEU-134	3.64		
Puerarin	ILE-10	4.09	ILE-10	3.74		
	THR-14	3.17	ALA-31	3.37		
	THR-14	3.97	GLN-131	3.77		
	LEU-83	2.67	ASN-132	3.75		
	ASP-86	3.62	LEU-134	3.67		
	ASP-86	3.12				
	LYS-89	3.46				
	GLN-131	3.03				

Table 2

AutoDock Vina binding energy of the 4KD1 receptor with the top 10 promising ligands

Receptor ID	**Ligand**		**Ligand**	
	CID	**Binding Energy**	**CID**	**Binding Energy**
4KD1	Hesperidin (10621)	-10.078	Eriocitrin (83489)	-9.325
4KD1	CorylifolA (25056407)	-9.569	Weighteone (5281814)	-9.240
4KD1	Catechin gallate (5276454)	-9.516	Lupiwighteone (5317480)	-9.112
4KD1	Baicalin (64982)	-9.400	Gallocatechin gallate (5276890)	-9.047
4KD1	Corylin (5316097)	-9.330	Puerarin (5281807)	-9.006

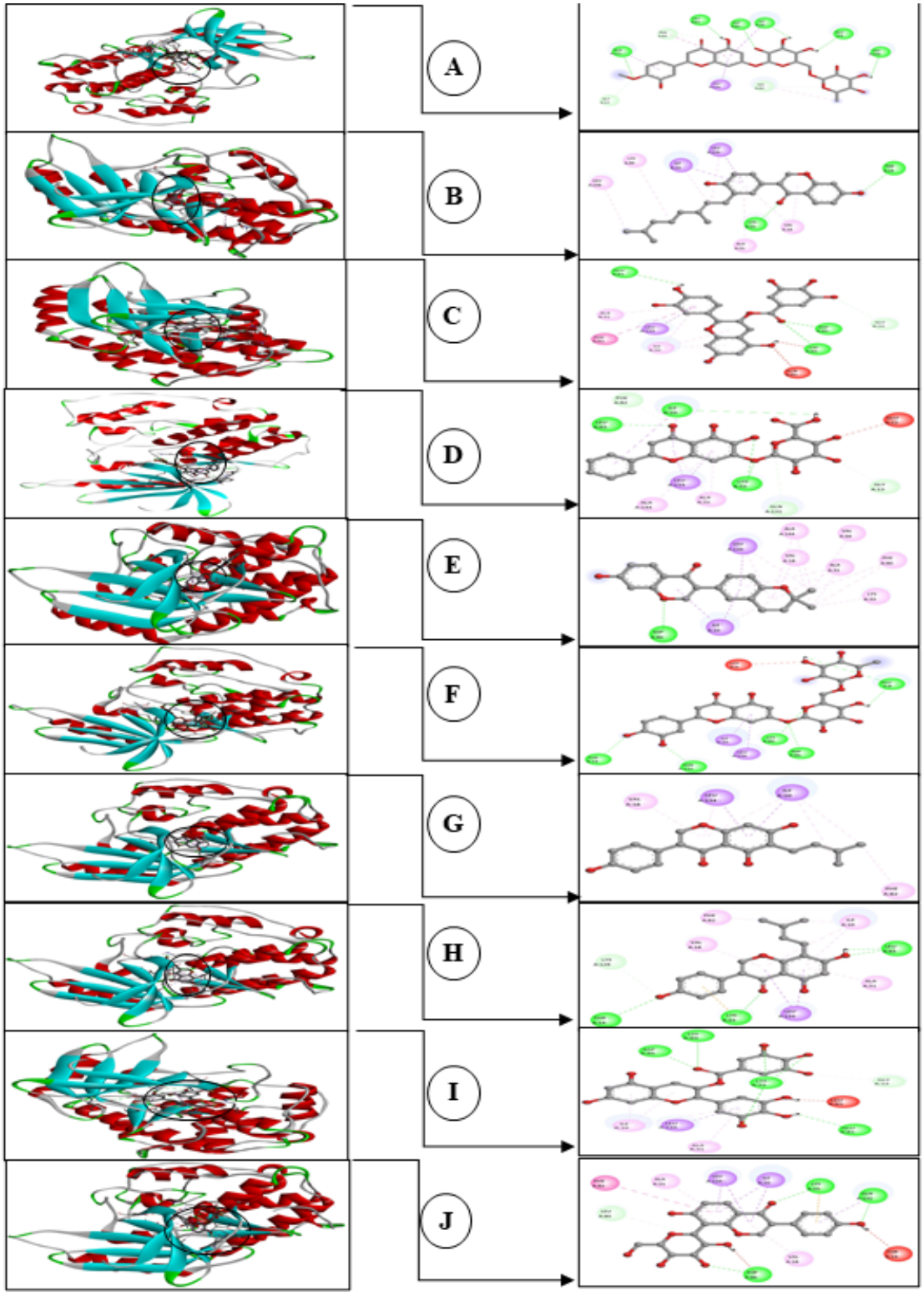


Fig. 1. Receptor-ligand complex and 2D structure of A. Hesperidin B. Corylifol A. Catechin gallate D. Baicalin E. Corylin F. Eriocitin G. Weighteone H. Lupiwighteone I. Gallocatechin gallate and J. Puerarin.

3.2 ADMET Analysis

The absorption, distribution, metabolism, excretion, and toxicity profiles of the ten best-docked compounds. Table 3 reports predicted LD50 values, toxicity classes, and prediction accuracy. Table 4 reports predicted inhibition profiles for CYP1A2, CYP2C19, CYP2C9, CYP2D6, and CYP3A4. Table 5 summarizes predicted gastrointestinal (GI) absorption, blood–brain barrier (BBB) permeability, hepatotoxicity, Ames mutagenicity, skin sensitization, and drug-likeness.

Table 3

The predicted LD50 using Protox 3.0

Compound Name	**Predicted Oral LD50 (mg/kg)**	**Toxicity Class**	**Prediction Accuracy (%)**
Hesperidin	12,000	6	72.9
CorylifolA	2,500	5	68.07
Catechin gallate	1,000	4	100
Baicalin	5,000	5	69.26
Corylin	3,850	5	69.26
Eriocitrin	12,000	6	72.9
Wighteone	2,500	5	69.26
Lupiwighteone	2,500	5	68.07
Gallocatechin gallate	1000	4	100
Puerarin	832	4	68.07

Table 4

CYP inhibition values from ADMETlab

Compound Name	**CYP1A2**	**CYP2C19**	**CYP2C9**	**CYP2D6**	**CYP3A4**
Hesperidin	$1.74x10^{-12}$	$3.95x10^{-7}$	$1.99x10^{-5}$	$7.42x10^{-11}$	$2.14x10^{-7}$
CorylifolA	0.850	1.000	0.995	0.929	0.931
Catechin gallate	$2.07x10^{-5}$	$8.82x10^{-5}$	0.0106	$4.35x10^{-7}$	0.761
Baicalin	0.0112	$6.57x10^{-6}$	$7.80x10^{-7}$	$1.61x10^{-5}$	$1.24x10^{-4}$
Corylin	0.992	1.000	0.920	0.956	0.393
Eriocitrin	$6.78x10^{-10}$	$5.86x10^{-4}$	$1.81x10^{-4}$	$1.98x10^{-9}$	0.00153
Wighteone	0.882	1.000	0.994	0.984	0.955
Lupiwighteone	0.241	1.000	1.000	0.955	0.275
Gallocatechin gallate	$1.57x10^{-6}$	$4.21x10^{-6}$	0.632	$3.35x10^{-9}$	0.925
Puerarin	0.0114	$7.40x10^{-5}$	0.0788	0.00703	0.00433

Table 5

ADMETLab results

Compound Name	GI Absorption (HIA)	BBB Permeability	Hepatotoxicity	Ames Mutagenicity	Skin sensitization	Drug likeness (QED)
Hesperidin	0.317	0.000020	0.631	0.944	0.983	0.185
CorylifolA	0.020	0.000014	0.715	0.264	0.829	0.498
Catechin gallate	0.001	0.007978	0.260	0.595	1.000	0.235
Baicalin	0.008	0.000993	0.429	0.500	0.966	0.299
Corylin	0.008	0.158	0.538	0.722	0.197	0.724
Eriocitrin	0.912	0.000238	0.720	0.916	0.997	0.724
Wighteone	0.033	0.022704	0.542	0.617	0.885	0.173
Lupiwighteone	0.010	0.006314	0.543	0.621	0.837	0.626
Gallocatechin gallate	0.003	0.001837	0.219	0.674	1.000	0.212
Puerarin	0.758	0.038980	0.669	0.786	0.963	0.355

3.3 Molecular dynamics simulation

A 50 ns simulation was run for the CDK2-Hesperidin complex; the root-mean-square (RMSD) appears to have remained relatively stable, fluctuating between 0.25 and 0.30nm, with no sustained increase, indicating overall protein stability. Although Hesperidin exhibited greater conformational variation, its RMSD appears to have fluctuated continuously at approximately 0.4–0.6 nm around 20 ns (Fig. 2).

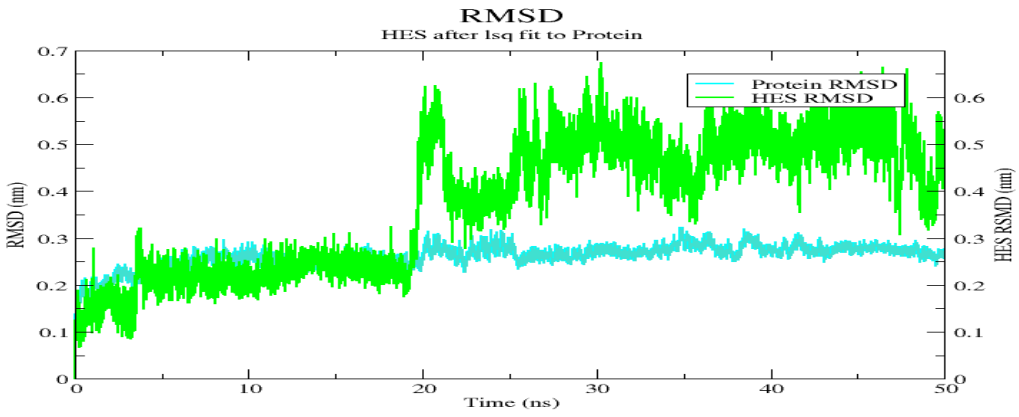


Fig. 2. Receptor (4KD1) Ligand Hesperidin RMSD plots.

Protein root-mean-square fluctuation (RMSF) ranged from 0.045 to 0.597 nm, peaking at residue 298 (Fig. 3). Protein conformation remained stable throughout the simulation.

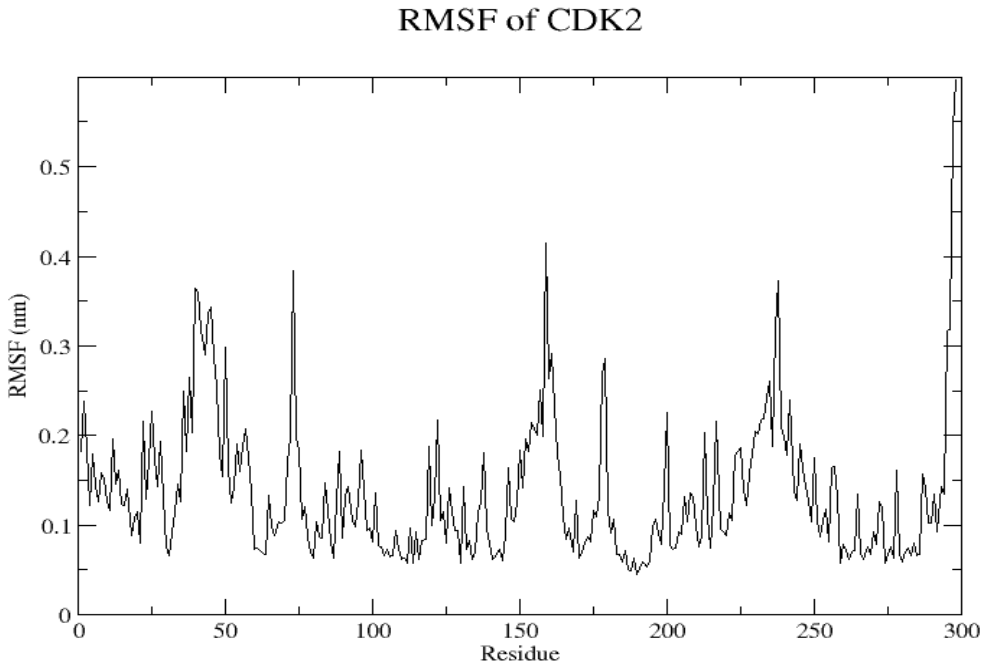


Fig. 3. Receptor (4KD1) Ligand Hesperidin RMSF plot.

Protein radius of gyration (Rg) varied from 1.99 to 2.08 nm (Fig. 4), demonstrating minimal change in protein compactness during the simulation.

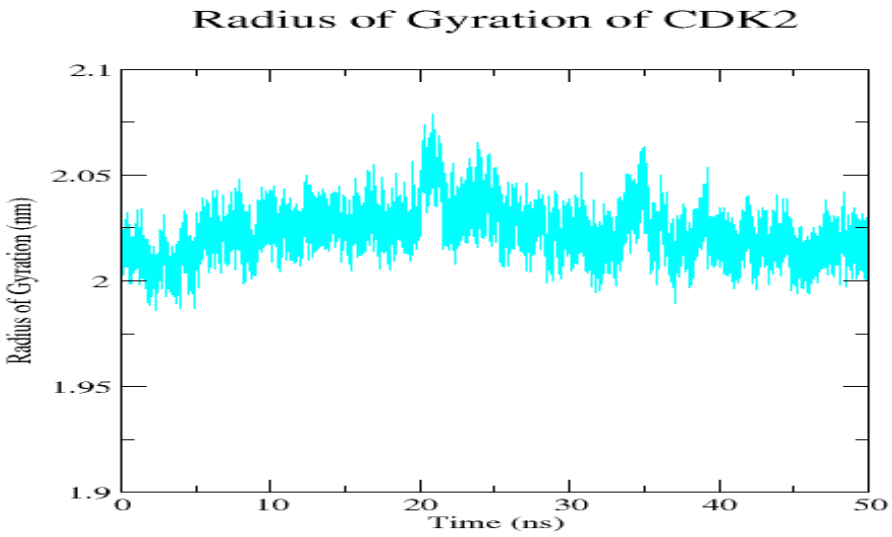


Fig. 4. Receptor (4KD1) Ligand Hesperidin Radius of gyration Rg plots.

Protein solvent-accessible surface area (SASA) ranged from 145.44 to 167.74 nm^2, with no pronounced increase or decrease. The protein–Hesperidin complex showed minimal energy fluctuation and no long-term drift (Fig. 5). The complex remained stable throughout the 50 ns simulation.

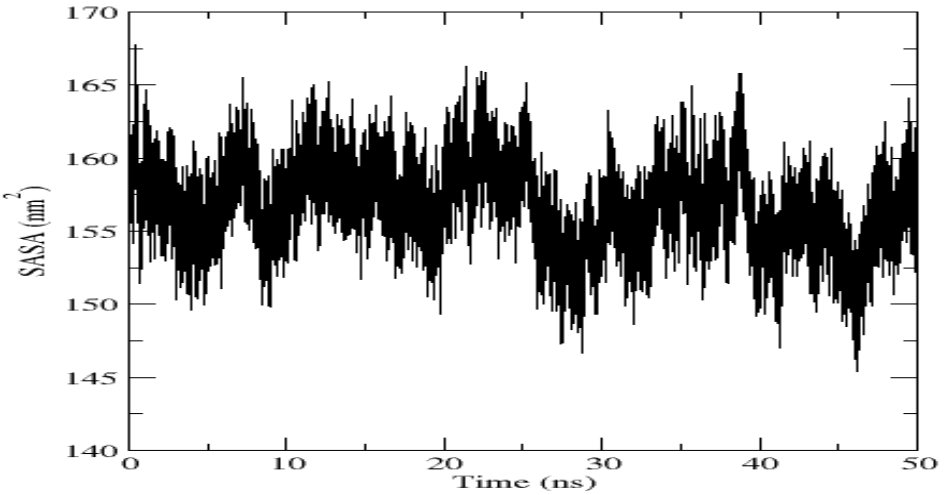


Fig. 5. Receptor (4KD1) ligand Hesperidin solvent accessible surface Area of the simulation result plot.

Contacts between the protein and hesperidin within 0.6 nm fluctuated, demonstrating dynamic residue-level interactions during the simulation (Fig. 6).

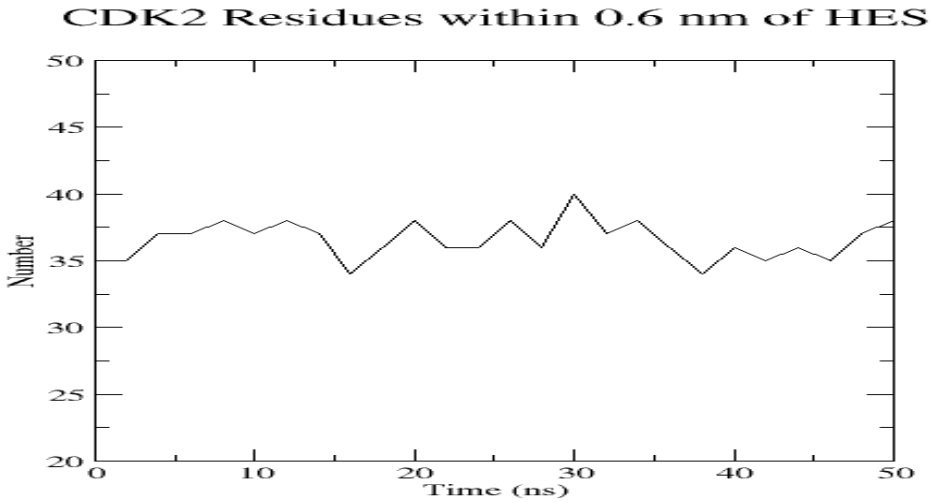


Fig. 6. Receptor (4KD1) ligand hesperidin Hydrogen contact plot.

The system temperature remained near 300 K throughout the simulation, with minor fluctuations (Fig. 7), demonstrating stable thermal conditions. By contrast, pressure fluctuated around a mean of 0.82 bar.

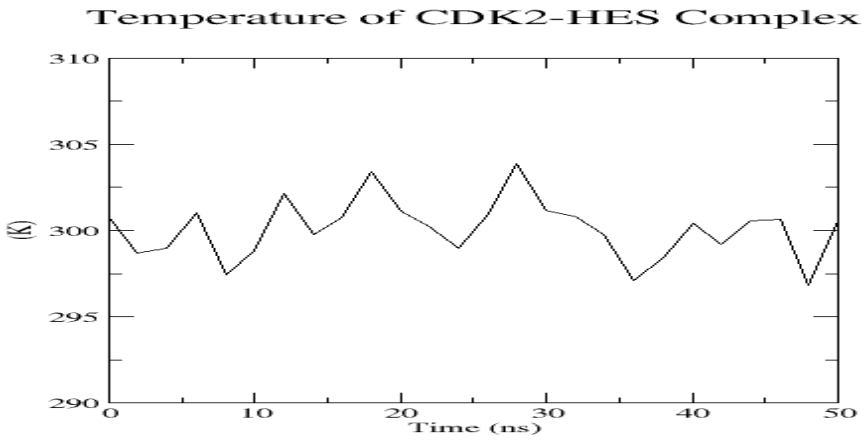


Fig. 6. Receptor (4KD1) Ligand Hesperidin complex plot.

The 20 ns molecular-dynamics simulation was also conducted to evaluate complex stability after the initial docking-score assessment. CDK2 Backbone RMSD: The protein backbone RMSD stabilized within approximately **1.5–2.3 Å**, with a relatively stable profile toward the latter part of the 20 ns trajectory. The CDK2 structure remained stable, with no large-scale conformational disruption. Corylifol A RMSD**:** After correcting for periodic box-wrapping artifacts, Corylifol A showed a relatively stable positional profile within the binding region, with RMSD values of approximately **2.5–3.1 Å**. These RMSD values demonstrate Corylifol A retention within the binding site throughout the trajectory. See Fig. 7.

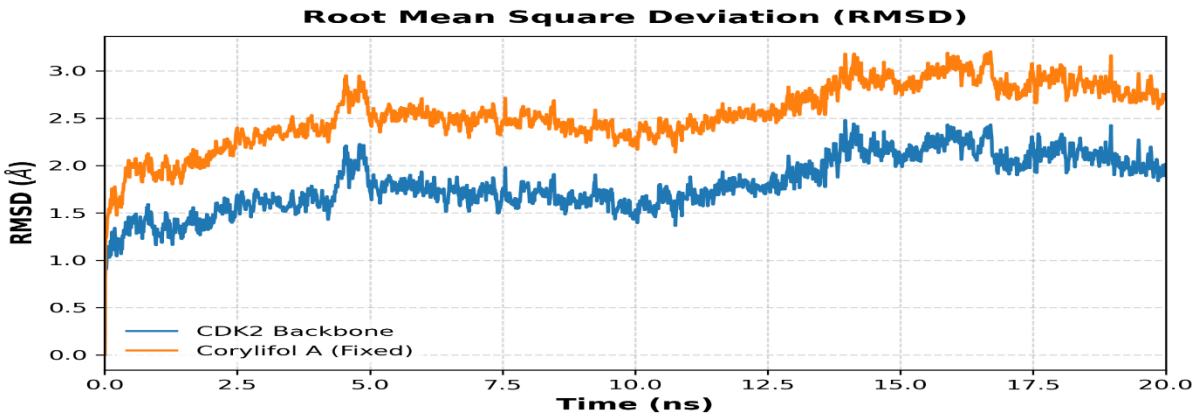


Fig. 2. Receptor (4KD1) Ligand CorylifolA reference RMSD plots.

Active-Site Residues**:** Residues surrounding the binding region generally exhibited low fluctuations, with root mean square fluctuation (**RMSF**) values below approximately **1.5 Å**. These values demonstrate limited local flexibility in the binding site during the simulation. Terminal Flexibility**:** The C-terminal residue 298 showed higher fluctuation, consistent with terminal mobility. This localized fluctuation alone does not establish global structural instability.

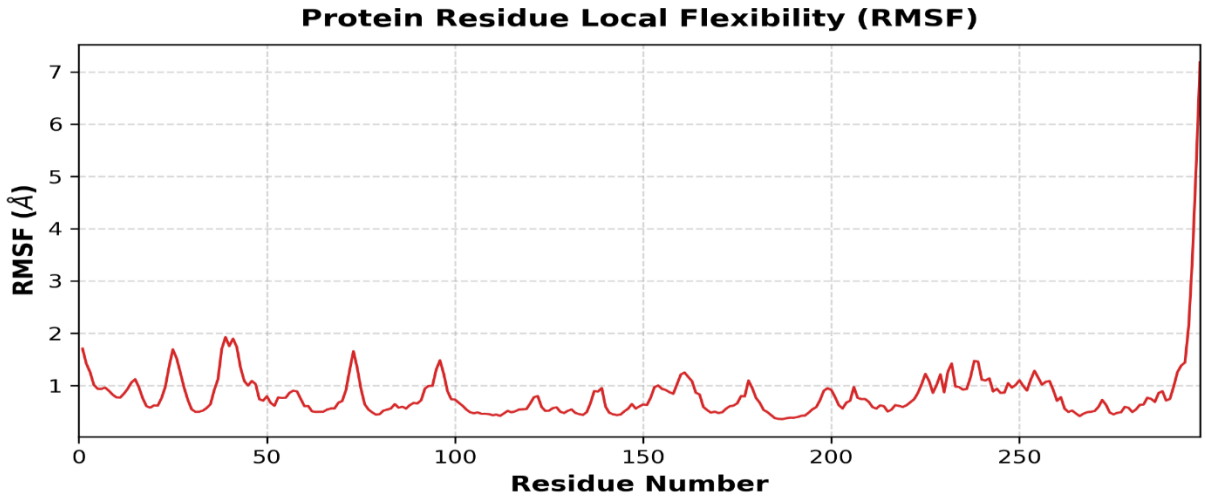


Fig. 7. Receptor (4KD1) Ligand CorylifolA RMSF plot.

Radius of Gyration (Rg): The radius of gyration remained relatively constant throughout the 20 ns trajectory, fluctuating between approximately **1.98 and 2.03 nm**. The stable Rg profile demonstrates that CDK2 maintained its compactness, with no major expansion or collapse. See Fig. 8.

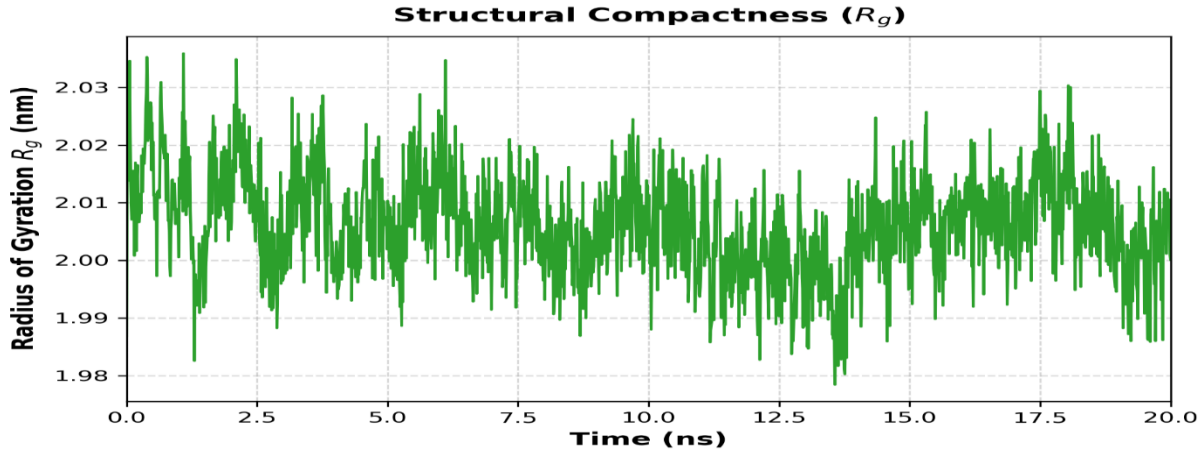


Fig. 8. Receptor (4KD1) Ligand CorylifolA Reference Radius of gyration Rg plots

The solvent-accessible surface area remained within approximately **150–163 nm²** without major systematic changes. This relatively stable SASA profile further supports maintenance of the overall structural conformation during the simulation. SASA alone does not establish specific shielding of hydrophobic residues by Corylifol A. see Fig. 9.

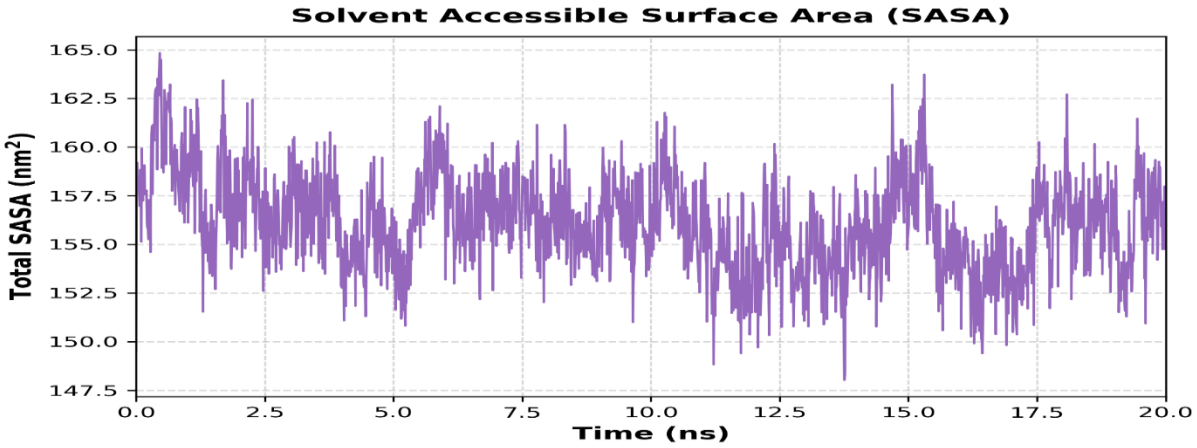


Fig. 9. Receptor (4KD1) ligand CorylifolA solvent accessible surface Area of the simulation result plot.

Hydrogen Bonds: The CDK2–Corylifol A complex maintained protein–ligand hydrogen bonds throughout the trajectory. The trajectory contained a mean of **1.5** hydrogen bonds and a maximum of 4 simultaneous hydrogen bonds. These interactions demonstrate persistent polar contacts between Corylifol A and binding-site residues. Key Residues**:** Trajectory-based interaction analysis identified **Leu83** and **Lys33** among the residues involved in ligand interactions. Repeated interactions identify Leu83 and Lys33 as contributors to Corylifol A stabilization in the CDK2 binding region. These interactions align with known ATP-binding-site interactions of CDK2 inhibitors but do not establish an identical binding mode. See Fig. 10.

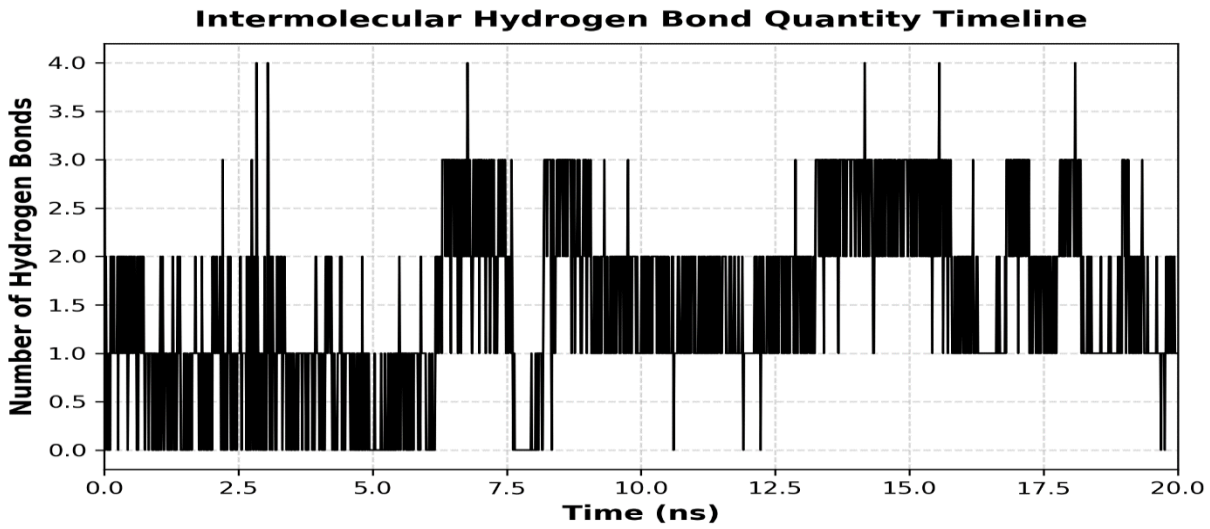


Fig. 10. Receptor (4KD1) CorylifolA complex hydrogen-bond plots.

MM/GBSA analysis of the CDK2–Corylifol A complex calculated a binding free energy (ΔGbind) of −43.63 ± 3.99 kcal/mol. The calculated gas-phase contribution (ΔGGAS) was −108.00 ± 14.83 kcal/mol, while the solvation contribution (ΔGSOLV) was +64.37 ± 12.14 kcal/mol. The van der Waals and electrostatic contributions were −49.48 ± 3.53 kcal/mol and −58.52 ± 14.52 kcal/mol, respectively. The polar GB solvation contribution (ΔEGB) was +70.21 ± 12.24 kcal/mol; the non-polar surface-area contribution (ΔESURF) was −5.84 ± 0.20 kcal/mol. We analyzed 101 snapshots from 15–20 ns of the molecular dynamics trajectory with gmx_MMPBSA v1.7.0 and the Generalized Born approach (igb = 2).

Energetic Contributions**:** The favorable estimate was strongly influenced by the electrostatic contribution (ΔEele = −58.52 kcal/mol) and van der Waals contribution (ΔEvdW = −49.48 kcal/mol). These favorable terms were partially opposed by the positive polar solvation/GB contribution (ΔEGB = +70.21 kcal/mol). Overall, the favorable molecular-mechanics interactions may have outweighed the unfavorable solvation contribution in the calculated MM/GBSA estimate.

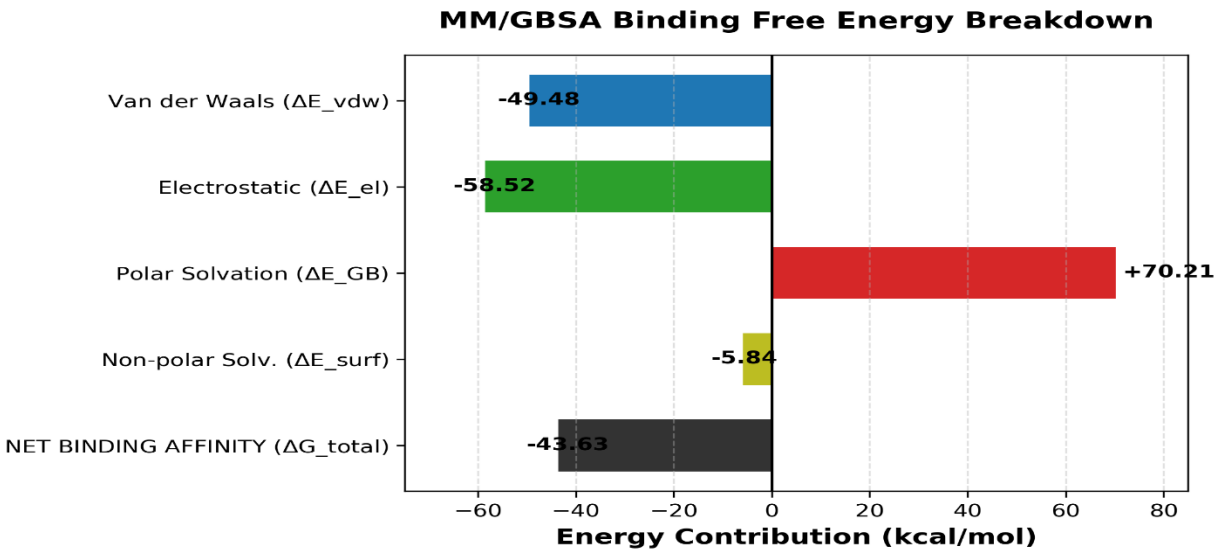


Fig. 11. Receptor (4KD1) CorylifolA Binding Energy plot

Together, the **RMSD, RMSF, Rg, SASA, hydrogen-bond, and MM/GBSA results** demonstrate structural stability in the CDK2–Hesperidin and CDK2–Corylifol A complexes across the 50 and 20 ns simulations. The simulations show a stable protein backbone, conserved compactness, limited active-site fluctuations, persistent protein–ligand hydrogen bonds, and favorable MM/GBSA energy. These results demonstrate stable CorylifolA binding to the CDK2 region during the simulated period.

Discussion

Flavonoid compounds are well-known plant phytochemicals and are used traditionally for certain diseases (Intharuksa et al., 2024). This study focused on the 100-compound flavonoid library. The results identify flavonoid candidates for experimental validation in drug development targeting cyclin-dependent kinase 2 (CDK2) using computer-aided drug design (Sony & Suresh, 2024). Molecular docking rapidly identifies receptor binding-site residues and estimates ligand-binding energies (Scotti et al., 2013). Recent studies show that flavonoids can bind to protein-interacting residues and serve as a new therapeutic method, and the computational work also helps reduce time and cost (Usha et al., 2018). We subjected 100 ligand compounds to virtual screening and molecular docking, followed by absorption, distribution, metabolism, excretion, and toxicity (ADMET) analysis and molecular dynamics simulation. We selected the ten docked compounds

with the highest binding energies as protein-binding candidates. Docking demonstrated favorable receptor-binding energies (kcal/mol): -10.078, -9.569, -9.516, -9.400, -9.330, -9.325, -9.112, -9.047, and -9.006, for Hesperidin, CorylifolA, Catechin gallate, Baicalin, Corylin, Eriocitrin, Weighteone, Lupiwighteone, Gallocatechin gallate, and Puerarin, respectively. This was also reported in closely related work (Sabe et al., 2021). These findings demonstrate that the compounds interact with the receptor protein's active site (Martin & Schonbrunn, 2013). ADMET analysis predicted LD50 values of 832–12,000 mg/kg and toxicity classes of 4–6 for the ten selected compounds. Hesperidin and Eriocitin showed the highest LD50 of 12,000mg/kg with a toxicity class of 6. Puerarin showed the smallest value of 832mg/kg with a toxicity class of 4. Catechin gallate and gallo catechin gallate were also assigned toxicity class 4; these findings suggest that compounds with higher LD50 values may exhibit lower toxicity. Some recent studies also followed this method (Bitew et al., 2021). The predicted cytochrome P450 (CYP) inhibition profiles of the selected compounds Hesperidin and Eriocitin indicated low inhibition across CYP isoforms. GI absorption reached 0.912 for Eriocitin and 0.758 for Puerarin. Most other compounds showed lower absorption; Corylin reached 0.158. Hepatotoxicity of the compounds varies from 0.219 to 0.720; Ames mutagenicity and skin sensitization showed differences among compounds and also ranged from 0.173 to 0.724; Corylin and Eriocitrin showed the highest values (S. B. Kim et al., 2014; Zheng et al., 2010)

We conducted two molecular dynamics (MD) simulations: the CDK2–Hesperidin complex for 50 ns and the CDK2–CorylifolA complex for 20 ns. During the 50 ns simulation, the protein RMSD profile suggested that the CDK2–hesperidin complex remained stable after initial equilibration, with values largely within approximately 0.25–0.30 nm and no subsequent increase. The RMSF profile suggested relatively low residue-level fluctuations, although flexibility appeared greater near the C-terminal residue. The radius of gyration also showed stability, and the SASA profile further indicated that the stability of the solvent exposure was maintained during the simulation. A previous study also analyzed these parameters (Ghahremanian et al., 2022). In addition, the protein–hesperidin contact profile suggested interactions with residues in the active-site binding region. The MM/GBSA analysis suggested a favorable binding energy of −40.89 kcal/mol for the CDK2–hesperidin complex during the 50 ns simulation. Furthermore, the 20 ns simulation suggested that the CDK2–CorylifolA complex remained stable, with protein RMSD values of

approximately 0.15–0.23 nm and ligand RMSD values of 0.25–0.31 nm; the CDK2 RMSF profile also suggested limited fluctuations in the binding region. Hydrogen bonding involving Leu83 and Lys33 may further contribute to CorylifolA stabilization within the binding region. The MM/GBSA estimate was −43.63 kcal/mol, with van der Waals and electrostatic interactions appearing to contribute substantially; this result appears broadly consistent with prior findings (Valdés-Tresanco et al., 2021).

Conclusion

We used molecular docking to evaluate flavonoid binding to CDK2, a component of the cancer cell-cycle machinery. The compounds showed favorable predicted binding affinities. Flavonoids may have therapeutic potential across a range of diseases. Among the docked compounds, selected candidates underwent further assessment of drug-likeness and stability over the simulation period. The compound's interactions with the protein may warrant investigation of its potential to address CDK2-associated drug resistance and its suitability for further development. Further investigation is warranted for candidates with the strongest predicted binding interactions. We recommend evaluating these candidates experimentally to assess their potential relevance to human health.

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References

Abraham, M. J., Murtola, T., Schulz, R., Páll, S., Smith, J. C., Hess, B., & Lindah, E. (2015). Gromacs: High performance molecular simulations through multi-level parallelism from laptops to supercomputers. *SoftwareX*, *1–2*, 19–25. https://doi.org/10.1016/j.softx.2015.06.001

Adasme, M. F., Linnemann, K. L., Bolz, S. N., Kaiser, F., Salentin, S., Haupt, V. J., & Schroeder, M. (2021). PLIP 2021: Expanding the scope of the protein-ligand interaction profiler to DNA and RNA. *Nucleic Acids Research*, *49*, W530–W534. https://doi.org/10.1093/nar/gkab294

Almalki, S. G. (2023). The pathophysiology of the cell cycle in cancer and treatment strategies using various cell cycle checkpoint inhibitors. In *Pathology Research and Practice* (Vol. 251). Elsevier GmbH. https://doi.org/10.1016/j.prp.2023.154854

Asghar, U., Witkiewicz, A. K., Turner, N. C., & Knudsen, E. S. (2015). The history and future of targeting cyclin-dependent kinases in cancer therapy. In *Nature Reviews Drug Discovery* (Vol. 14, pp. 130–146). Nature Publishing Group. https://doi.org/10.1038/nrd4504

Bach, S., Knockaert, M., Reinhardt, J., Lozach, O., Schmitt, S., Baratte, B., Koken, M., Coburn, S. P., Tang, L., Jiang, T., Liang, D. C., Galons, H., Dierick, J. F., Pinna, L. A., Meggio, F., Totzke, F., Schächtele, C., Lerman, A. S., Carnero, A., … Meijer, L. (2005). Roscovitine targets, protein kinases and pyridoxal kinase. *Journal of Biological Chemistry*, *280*, 31208–31219. https://doi.org/10.1074/jbc.M500806200

Banerjee, P., Kemmler, E., Dunkel, M., & Preissner, R. (2024). ProTox 3.0: A webserver for the prediction of toxicity of chemicals. *Nucleic Acids Research*, *52*, W513–W520. https://doi.org/10.1093/nar/gkae303

Benson, C., White, J., De Bono, J., O'Donnell, A., Raynaud, F., Cruickshank, C., McGrath, H., Walton, M., Workman, P., Kaye, S., Cassidy, J., Gianella-Borradori, A., Judson, I., & Twelves, C. (2007). A phase I trial of the selective oral cyclin-dependent kinase inhibitor seliciclib (CYC202; R-Roscovitine), administered twice daily for 7 days every 21 days. *British Journal of Cancer*, *96*, 29–37. https://doi.org/10.1038/sj.bjc.6603509

Bitew, M., Desalegn, T., Demissie, T. B., Belayneh, A., Endale, M., & Eswaramoorthy, R. (2021). Pharmacokinetics and drug-likeness of antidiabetic flavonoids: Molecular docking and DFT study. *PLoS ONE*, *16*. https://doi.org/10.1371/journal.pone.0260853

Bolton, E. E., Kim, S., & Bryant, S. H. (2011). PubChem3D: Conformer generation. *Journal of Cheminformatics*, *3*, 4. https://doi.org/10.1186/1758-2946-3-4

Bongrand, P. (1999). Ligand-receptor interactions. *Reports on Progress in Physics*, *62*, 921–968. https://doi.org/10.1088/0034-4885/62/6/202

Carlson, B. A., Dubay, M. M., Sausville, E. A., Brizuela, L., & Worland, P. J. (1996). Flavopiridol induces G1 arrest with inhibition of cyclin-dependent kinase (CDK) 2 and CDK4 in human breast carcinoma cells. *Cancer Research*, *56*, 2973–2978.

Chao, S. H., & Price, D. H. (2001). Flavopiridol Inactivates P-TEFb and Blocks Most RNA

Polymerase II Transcription in Vivo. *Journal of Biological Chemistry*, *276*, 31793–31799. https://doi.org/10.1074/jbc.M102306200

Davies, T. G., Pratt, D. J., Endicott, J. A., Johnson, L. N., & Noble, M. E. M. (2002). Structure-based design of cyclin-dependent kinase inhibitors. *Pharmacology and Therapeutics*, *93*, 125–133. https://doi.org/10.1016/S0163-7258(02)00182-1

De Azevedo, W. F., Mueller-Dieckmann, H. J., Schulze-Gahmen, U., Worland, P. J., Sausville, E., & Kim, S. H. (1996). Structural basis for specificity and potency of a flavonoid inhibitor of human CDK2, a cell cycle kinase. *Proceedings of the National Academy of Sciences of the United States of America*, *93*, 2735–2740. https://doi.org/10.1073/pnas.93.7.2735

Eberhardt, J., Santos-Martins, D., Tillack, A. F., & Forli, S. (2021). AutoDock Vina 1.2.0: New Docking Methods, Expanded Force Field, and Python Bindings. *Journal of Chemical Information and Modeling*, *61*, 3891–3898. https://doi.org/10.1021/acs.jcim.1c00203

Elmasry, G. F. (2023). Cyclin-dependent kinase 2 (CDK2). In *Metalloenzymes: From Bench to Bedside* (pp. 51–62). Elsevier. https://doi.org/10.1016/B978-0-12-823974-2.00030-9

Fry, D. W., Harvey, P. J., Keller, P. R., Elliott, W. L., Meade, M. A., Trachet, E., Albassam, M., Zheng, X. X., Leopold, W. R., Pryer, N. K., & Toogood, P. L. (2004). Specific inhibition of cyclin-dependent kinase 4/6 by PD 0332991 and associated antitumor activity in human tumor xenografts. In *Molecular Cancer Therapeutics* (Vol. 3, pp. 1427–1437). https://doi.org/10.1158/1535-7163.1427.3.11

Fu, L., Shi, S., Yi, J., Wang, N., He, Y., Wu, Z., Peng, J., Deng, Y., Wang, W., Wu, C., Lyu, A., Zeng, X., Zhao, W., Hou, T., & Cao, D. (2024). ADMETlab 3.0: an updated comprehensive online ADMET prediction platform enhanced with broader coverage, improved performance, API functionality and decision support. *Nucleic Acids Research*, *52*, W422–W431. https://doi.org/10.1093/nar/gkae236

Ghahremanian, S., Rashidi, M. M., Raeisi, K., & Toghraie, D. (2022). Molecular dynamics simulation approach for discovering potential inhibitors against SARS-CoV-2: A structural review. *Journal of Molecular Liquids*, *354*, 118901. https://doi.org/10.1016/j.molliq.2022.118901

House, I., Valore-Caplan, M., Maris, E., & Falchook, G. S. (2025). Cyclin Dependent Kinase 2 (CDK2) Inhibitors in Oncology Clinical Trials: A Review. *Journal of Immunotherapy and Precision Oncology*, *8*, 47–54. https://doi.org/10.36401/JIPO-24-22

Hsu, C. Y., Almajidi, Y. Q., Al-hakeem, M. abdulrazzaq, Alshahrani, M. Y., Nabil, W., Jayakumar, S. S., SinglaI, S., Al-Khafaji, Z. A., Hussein, A. R., & Al-Mashhadani, Z. I. (2025). Deregulated cell cycle control: The interplay between non-coding RNAs and cyclin-dependent kinases in tumorigenesis. In *Seminars in Oncology* (Vol. 52). W.B. Saunders. https://doi.org/10.1016/j.seminoncol.2025.152395

Hussein, H. A., Borrel, A., Geneix, C., Petitjean, M., Regad, L., & Camproux, A. C. (2015). PockDrug-Server: A new web server for predicting pocket druggability on holo and apo proteins. *Nucleic Acids Research*, *43*, W436–W442. https://doi.org/10.1093/nar/gkv462

Intharuksa, A., Kuljarusnont, S., Sasaki, Y., & Tungmunnithum, D. (2024). Flavonoids and

Other Polyphenols: Bioactive Molecules from Traditional Medicine Recipes/Medicinal Plants and Their Potential for Phytopharmaceutical and Medical Application. In *Molecules* (Vol. 29). Multidisciplinary Digital Publishing Institute (MDPI). https://doi.org/10.3390/molecules29235760

Janave, M. T., Ramaswamy, N. K., & Nair, P. M. (1999). Studies on determination of active site amino acid residues in glyoxylate synthetase from potato tuber chloroplasts. *Plant Physiology and Biochemistry*, *37*, 121–129. https://doi.org/10.1016/S0981-9428(99)80073-4

Jorgensen, W. L., Chandrasekhar, J., Madura, J. D., Impey, R. W., & Klein, M. L. (1983). Comparison of simple potential functions for simulating liquid water. *The Journal of Chemical Physics*, *79*, 926–935. https://doi.org/10.1063/1.445869

Kaur, G., Stetler-stevenson, M., Sebers, S., Worland, P., Sedlacek, H., Myers, C., Czech, J., Naik, R., & Sausville, E. (1992). Growth inhibition with reversible cell cycle arrest of carcinoma cells by flavone l86-8275. *Journal of the National Cancer Institute*, *84*, 1736–1740. https://doi.org/10.1093/jnci/84.22.1736

Keyomarsi, K., Tucker, S. L., Buchholz, T. A., Callister, M., Ding, Y., Hortobagyi, G. N., Bedrosian, I., Knickerbocker, C., Toyofuku, W., Lowe, M., Herliczek, T. W., & Bacus, S. S. (2002). Cyclin E and Survival in Patients with Breast Cancer. *New England Journal of Medicine*, *347*, 1566–1575. https://doi.org/10.1056/nejmoa021153

Kim, S. B., Yoon, I. S., Kim, K. S., Cho, S. J., Kim, Y. S., Cho, H. J., Chung, S. J., Chong, S., & Kim, D. D. (2014). In vitro and in vivo evaluation of the effect of puerarin on hepatic cytochrome P450-mediated drug metabolism. *Planta Medica*, *80*, 561–567. https://doi.org/10.1055/s-0034-1368350

Kim, S., Thiessen, P. A., Cheng, T., Yu, B., & Bolton, E. E. (2018). An update on PUG-REST: RESTful interface for programmatic access to PubChem. *Nucleic Acids Research*, *46*, W563–W570. https://doi.org/10.1093/nar/gky294

Le Tourneau, C., Faivre, S., Laurence, V., Delbaldo, C., Vera, K., Girre, V., Chiao, J., Armour, S., Frame, S., Green, S. R., Gianella-Borradori, A., Diéras, V., & Raymond, E. (2010). Phase i evaluation of seliciclib (R-roscovitine), a novel oral cyclin-dependent kinase inhibitor, in patients with advanced malignancies. *European Journal of Cancer*, *46*, 3243–3250. https://doi.org/10.1016/j.ejca.2010.08.001

Lindorff-Larsen, K., Piana, S., Palmo, K., Maragakis, P., Klepeis, J. L., Dror, R. O., & Shaw, D. E. (2010). Improved side-chain torsion potentials for the Amber ff99SB protein force field. *Proteins: Structure, Function and Bioinformatics*, *78*, 1950–1958. https://doi.org/10.1002/prot.22711

Lipinski, C. A., Lombardo, F., Dominy, B. W., & Feeney, P. J. (1997). Experimental and computational approaches to estimate solubility and permeability in drug discovery and development settings. In *Advanced Drug Delivery Reviews* (Vol. 23, pp. 3–25). Elsevier Science B.V. https://doi.org/10.1016/S0169-409X(96)00423-1

Losiewicz, M. D., Carlson, B. A., Kaur, G., Sausville, E. A., & Worland, P. J. (1994). Potent inhibition of Cdc2 kinase activity by the flavonoid L86-8275. *Biochemical and Biophysical*

Research Communications, *201*, 589–595. https://doi.org/10.1006/bbrc.1994.1742

Martin, M. P., & Schonbrunn, E. (2013). CDK2 in complex with Dinaciclib. In *Worldwide Protein Data Bank*. https://doi.org/10.2210/pdb4kd1/pdb

Meijer, L., Borgne, A., Mulner, O., Chong, J. P. J., Blow, J. J., Inagaki, N., Inagaki, M., Delcros, J. G., & Moulinoux, J. P. (1997). Biochemical and cellular effects of roscovitine, a potent and selective inhibitor of the cyclin-dependent kinases cdc2, cdk2 and cdk5. *European Journal of Biochemistry*, *243*, 527–536. https://doi.org/10.1111/j.1432-1033.1997.t01-2-00527.x

O'Boyle, N. M., Banck, M., James, C. A., Morley, C., Vandermeersch, T., & Hutchison, G. R. (2011). Open Babel: An Open chemical toolbox. *Journal of Cheminformatics*, *3*. https://doi.org/10.1186/1758-2946-3-33

Rose, P. W., Prlić, A., Altunkaya, A., Bi, C., Bradley, A. R., Christie, C. H., Di Costanzo, L., Duarte, J. M., Dutta, S., Feng, Z., Green, R. K., Goodsell, D. S., Hudson, B., Kalro, T., Lowe, R., Peisach, E., Randle, C., Rose, A. S., Shao, C., … Burley, S. K. (2017). The RCSB protein data bank: Integrative view of protein, gene and 3D structural information. *Nucleic Acids Research*, *45*, D271–D281.

Sabe, V. T., Ntombela, T., Jhamba, L. A., Maguire, G. E. M., Govender, T., Naicker, T., & Kruger, H. G. (2021). Current trends in computer aided drug design and a highlight of drugs discovered via computational techniques: A review. In *European Journal of Medicinal Chemistry* (Vol. 224). Elsevier Masson s.r.l. https://doi.org/10.1016/j.ejmech.2021.113705

Salentin, S., Schreiber, S., Haupt, V. J., Adasme, M. F., & Schroeder, M. (2015). PLIP: Fully automated protein-ligand interaction profiler. *Nucleic Acids Research*, *43*, W443–W447. https://doi.org/10.1093/nar/gkv315

Scotti, L., Jaime Bezerra Mendonca Junior, F., Rodrigo Magalhaes Moreira, D., Sobral da Silva, M., R. Pitta, I., & Tullius Scotti, M. (2013). SAR, QSAR and Docking of Anticancer Flavonoids and Variants: A Review. *Current Topics in Medicinal Chemistry*, *12*, 2785–2809. https://doi.org/10.2174/1568026611212240007

Senderowicz, A. M. (1999). Flavopiridol: The first cyclin-dependent kinase inhibitor in human clinical trials. *Investigational New Drugs*, *17*, 313–320. https://doi.org/10.1023/A:1006353008903

Senderowicz, A. M., Headlee, D., Stinson, S. F., Lush, R. M., Kalil, N., Villalba, L., Hill, K., Steinberg, S. M., Figg, W. D., Tompkins, A., Arbuck, S. G., & Sausville, E. A. (1998). Phase I trial of continuous infusion flavopiridol, a novel cyclin- dependent kinase inhibitor, in patients with refractory neoplasms. *Journal of Clinical Oncology*, *16*, 2986–2999. https://doi.org/10.1200/JCO.1998.16.9.2986

Siddique, R., Thangavelu, L., S, R. J., Almalki, W. H., Kazmi, I., Kumar, A., Mahajan, S., Kalra, H., Alzarea, S. I., & Pant, K. (2024). lncRNAs and cyclin-dependent kinases: Unveiling their critical roles in cancer progression. In *Pathology Research and Practice* (Vol. 258). Elsevier GmbH. https://doi.org/10.1016/j.prp.2024.155333

Sony, A. S., & Suresh, M. X. (2024). Docking and Molecular Dynamics Studies on Anticancer

Activities of Flavonoids as Inhibitors of CDK2 and CDK9. *Medicinal Chemistry*, *21*, 69–83. https://doi.org/10.2174/0115734064314933240812120123

Sousa Da Silva, A. W., & Vranken, W. F. (2012). ACPYPE - AnteChamber PYthon Parser interfacE. *BMC Research Notes*, *5*. https://doi.org/10.1186/1756-0500-5-367

Stachyra, P., Hernando Meliá, C., López-Guerrero, J. A., Gavilá Gregori, J., & Guerrero-Zotano, Á. L. (2026). Erratum: From CDK4/6 to CDK2 and KAT: Evolving therapeutic approaches in HR+/HER2– breast cancer (The Breast (2026) 88, (S0960977626001153), (10.1016/j.breast.2026.104805)). In *Breast* (Vol. 88). Churchill Livingstone. https://doi.org/10.1016/j.breast.2026.104806

T101-2024 Notification regarding BIOVIA Discovery Studio 2025. (n.d.). Https://Www.3ds.Com/Support/Documentation/Resource-Library/T101-2024-Notification-Regarding-Biovia-Discovery-Studio-2025?Utm_source=chatgpt.Com.

Taheri, M., Shirvani-Farsani, Z., Hussen, B. M., Abak, A., Liehr, T., Ghafouri-Fard, S., & Baniahmad, A. (2026). Involvement of cyclins in the pathogenesis of cancer. *Journal of Advanced Research*. https://doi.org/10.1016/j.jare.2026.06.030

Tavsan, Z., & Kayali, H. A. (2019). Flavonoids showed anticancer effects on the ovarian cancer cells: Involvement of reactive oxygen species, apoptosis, cell cycle and invasion. *Biomedicine and Pharmacotherapy*, *116*. https://doi.org/10.1016/j.biopha.2019.109004

Thilakarathna, S. H., & Vasantha Rupasinghe, H. P. (2013). Flavonoid bioavailability and attempts for bioavailability enhancement. In *Nutrients* (Vol. 5, pp. 3367–3387). MDPI AG. https://doi.org/10.3390/nu5093367

Trott, O., & Olson, A. J. (2010). AutoDock Vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of Computational Chemistry*, *31*, 455–461. https://doi.org/10.1002/jcc.21334

Usha, T., Shanmugarajan, D., Goyal, A. K., Kumar, C. S., & Middha, S. K. (2018). Recent Updates on Computer-aided Drug Discovery: Time for a Paradigm Shift. *Current Topics in Medicinal Chemistry*, *17*, 3296–3307. https://doi.org/10.2174/1568026618666180101163651

Valdés-Tresanco, M. S., Valdés-Tresanco, M. E., Valiente, P. A., & Moreno, E. (2021). Gmx_MMPBSA: A New Tool to Perform End-State Free Energy Calculations with GROMACS. *Journal of Chemical Theory and Computation*, *17*, 6281–6291. https://doi.org/10.1021/acs.jctc.1c00645

Wang, Y., Huang, M., Zhou, X., Li, H., Ma, X., & Sun, C. (2025). Potential of natural flavonoids to target breast cancer angiogenesis (review). In *British Journal of Pharmacology* (Vol. 182, pp. 2235–2258). John Wiley and Sons Inc. https://doi.org/10.1111/bph.16275

Whittaker, S. R., Mallinger, A., Workman, P., & Clarke, P. A. (2017). Inhibitors of cyclin-dependent kinases as cancer therapeutics. In *Pharmacology and Therapeutics* (Vol. 173, pp. 83–105). Elsevier Inc. https://doi.org/10.1016/j.pharmthera.2017.02.008

Woyach, J. A., Lozanski, G., Ruppert, A. S., Lozanski, A., Blum, K. A., Jones, J. A., Flynn, J. M., Johnson, A. J., Grever, M. R., Heerema, N. A., & Byrd, J. C. (2012). Outcome of

patients with relapsed or refractory chronic lymphocytic leukemia treated with flavopiridol: Impact of genetic features. In *Leukemia* (Vol. 26, pp. 1442–1444). https://doi.org/10.1038/leu.2011.375

Zheng, J., Chen, B., Jiang, B., Zeng, L., Tang, Z. R., Fan, L., & Zhou, H. H. (2010). The effects of puerarin on CYP2D6 and CYP1A2 activities in vivo. *Archives of Pharmacal Research*, *33*, 243–246. https://doi.org/10.1007/s12272-010-0209-2